

Student

Axel Sneessens

Total Points

20.5 / 31 pts

Question 1

Selection and social influence in networks

0 / 5 pts

1.1 Definition of selection and social influence

0 / 1 pt

– 0 pts Correct

✓ – 1 pt Incorrect

– 0.5 pts Selection definition wrong

– 0.5 pts Social influence definition wrong

1.2 Wikipedia as a social network

0 / 1 pt

– 0 pts Correct

✓ – 1 pt Incorrect

– 0.5 pts Similarity formula defined by overlap in set of edited articles

– 0.5 pts Link defined by participation on the other's talk page

1.3 Comparing selection and social influence on Wikipedia

0 / 3 pts

– 0 pts Correct

✓ – 3 pts Incorrect

– 0.5 pts Selection explanation incomplete

– 1 pt Selection explanation missing

– 0.5 pts Social influence explanation incomplete

– 1 pt Social influence explanation missing

– 0.5 pts Speed explanation incomplete

– 1 pt Speed explanation missing or wrong

Question 2

Second-price auction theorem

0 / 5 pts

2.1 Second-price auction in the language of game theory

0 / 2 pts

– 0 pts Correct

✓ – 2 pts Incorrect

– 0.5 pts Payoff for losing bidder missing or wrong

– 1 pt Payoff for first-price given

– 1.5 pts Payoff for winner is wrong

– 0.5 pts Explanation incomplete or small error

2.2 Second-price auction theorem

0 / 3 pts

– 0 pts Correct

✓ – 3 pts Incorrect

– 2 pts Most of explanation missing

– 1.5 pts Reasoning errors

– 1 pt Part of explanation missing

– 0.5 pts Small errors

Question 3

Negotiations on networks

6.5 / 7 pts

3.1 The three-node path

1 / 1 pt

✓ - 0 pts Correct

- 1 pt Incorrect

- 0.5 pts Some explanation missing

3.2 The three principles of network negotiation

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Incorrect

- 0.5 pts Some explanation missing

- 0.5 pts Small error

- 1 pt Partly correct but with error

- 1.5 pts Correct but highly incomplete

3.3 Stable outcomes

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Incorrect

- 1.5 pts Some correct information

- 0.5 pts Definition incomplete

- 1 pt Definition incorrect

- 0.5 pts Example incomplete

- 1 pt Example incorrect

3.4 Balanced outcomes

1.5 / 2 pts

- 0 pts Correct

- 2 pts Incorrect

- 0.5 pts Definition incomplete

- 1 pt Definition incorrect

✓ - 0.5 pts Example incorrect but plausible

- 1 pt Example incorrect

Question 4

Bow-tie structure of the Web

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Incorrect
- 1 pt In and out parts not explained
- 1 pt Relative sizes missing or incomplete
- 1 pt Some explanation missing
- 1 pt Some errors

Question 5

Bayes' Rule

4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect
- 0.5 pts Small error
- 1 pt Calculation errors or missing formula
- 2 pts Partly correct but incomplete or errors in formula
- 3 pts Mostly incorrect

Question 6

Direct-benefit effects

7 / 7 pts

6.1 Market without direct-benefit effects

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Incorrect

- 1 pt Graph incomplete

- 1.5 pts Graph missing or wrong

- 1 pt Stability explanation incomplete

- 1.5 pts Stability explanation incorrect

6.2 Market with direct-benefit effects

4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect

- 0.5 pts New equilibrium price incomplete

- 1 pt New equilibrium price wrong or missing

- 0.5 pts Assumptions on functions incomplete

- 1 pt Assumptions on functions wrong or missing

- 0.5 pts Graph incomplete

- 1 pt Graph missing

- 0.5 pts Equilibria incomplete

- 1 pt Equilibria missing

LINFO1115 Final Exam

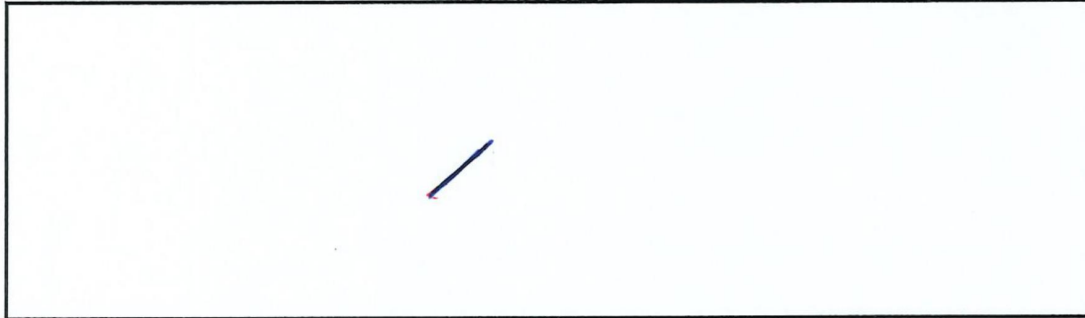
June 20, 2024

(Note: Q1 & Q2 taken together correspond to the midterm.)

Q1 Selection and social influence in social networks [5pts]

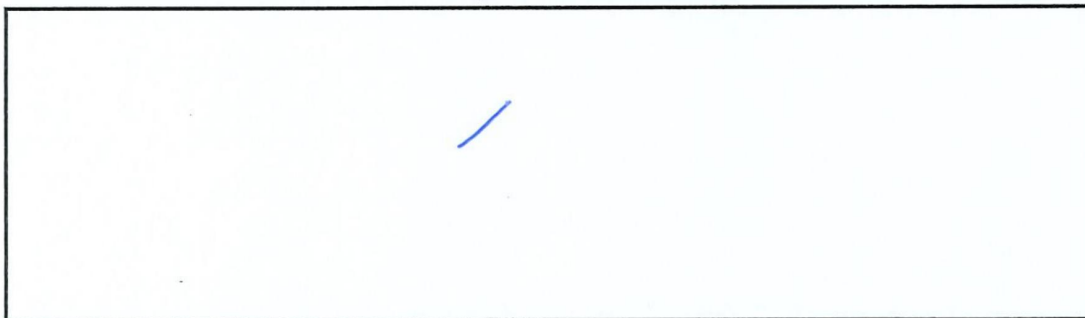
Q1.1 Definition of selection and social influence [1pt]

Define the concepts of selection and social influence in social networks.



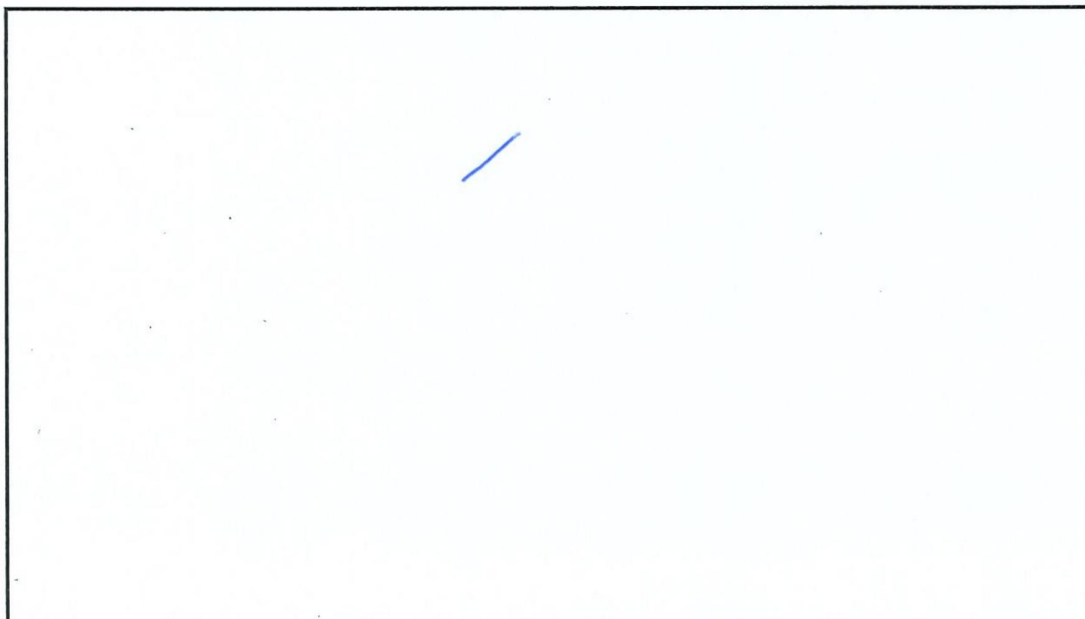
Q1.2 Wikipedia as a social network [1pt]

Selection and social influence have been compared in a Wikipedia study. Give a brief definition of Wikipedia as a social network where the participants are the editors. What do editors do? When are two editors linked? Define the numerical similarity between two editors.



Q1.3 Comparing selection and social influence on Wikipedia [3pts]

Explain how we compare selection and social influence on Wikipedia. Show the graph of average similarity and explain what it means in terms of selection and social influence. How do we distinguish between selection and social influence? How fast does similarity evolve in each case?



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Q2 Second-price auction theorem [5pts]

Q2.1 Second-price auction in the language of game theory [2pts]

Assume each bidder i is a player and bidder i 's strategy is the bid amount b_i as function of i 's true value v_i . A game is played when each bidder bids once; one bidder wins and the others lose. In your answer, define the *payoff* to each bidder i .

Q2.2 Second-price auction theorem [3pts]

We proved the theorem that in a sealed second-price auction, it is a *dominant strategy* for each bidder to choose a bid b_i equal to the bidder's true value v_i . To prove the theorem, we assume that each bidder bids $b_i = v_i$ and we show that no deviation from this will improve the payoff. The proof considers two cases: (1) i raises bid, and (2) i lowers bid. For this question, choose one of these two cases and explain precisely why i 's payoff will not improve in that case. You can make a diagram, as done in the course.

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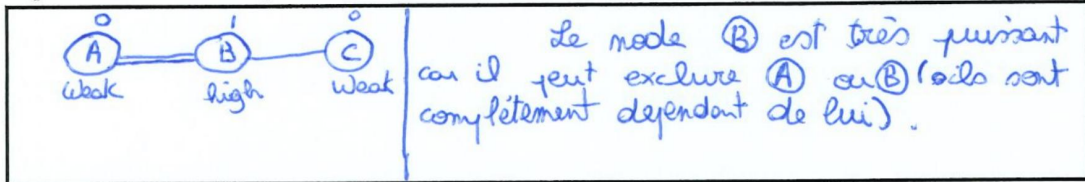
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Q3 Negotiations on networks [7pts]

Q3.1 The three-node path [1pt]

Explain the result of a negotiation on a three-node path. Which node(s) have the most power?

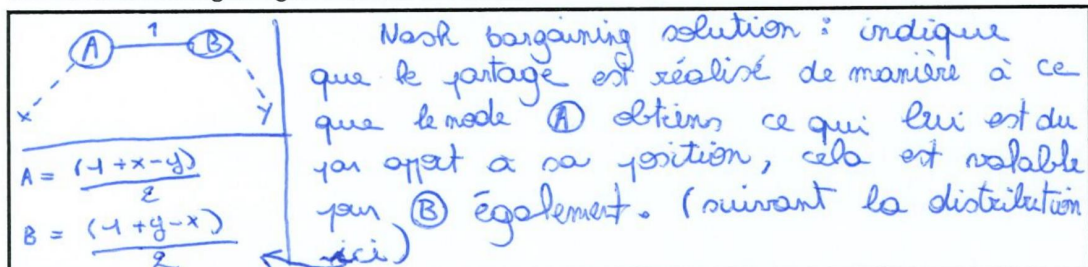


Q3.2 The three principles of network negotiation [2pt]

There are three important principles of network negotiation:

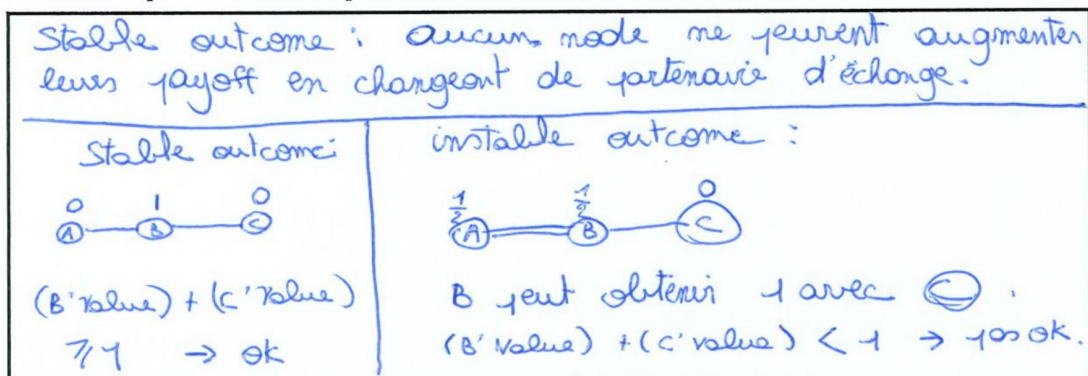
- Nash bargaining solution
- Ultimatum game
- Stable outcomes

Define the Nash bargaining solution for two nodes.



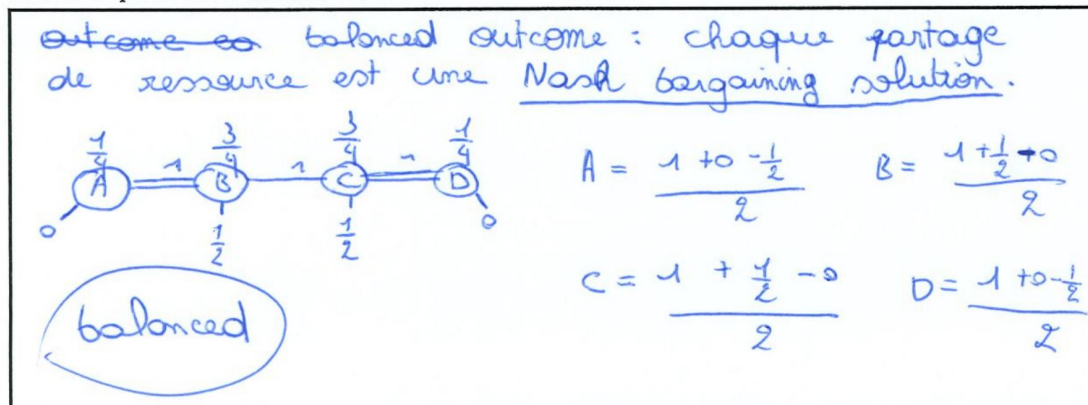
Q3.3 Stable outcomes [2pts]

Define the concept of stable outcome of a network exchange. Give an example of an unstable outcome in a three-node path. Give an example of a stable outcome in a three-node path.



Q3.4 Balanced outcomes [2pts]

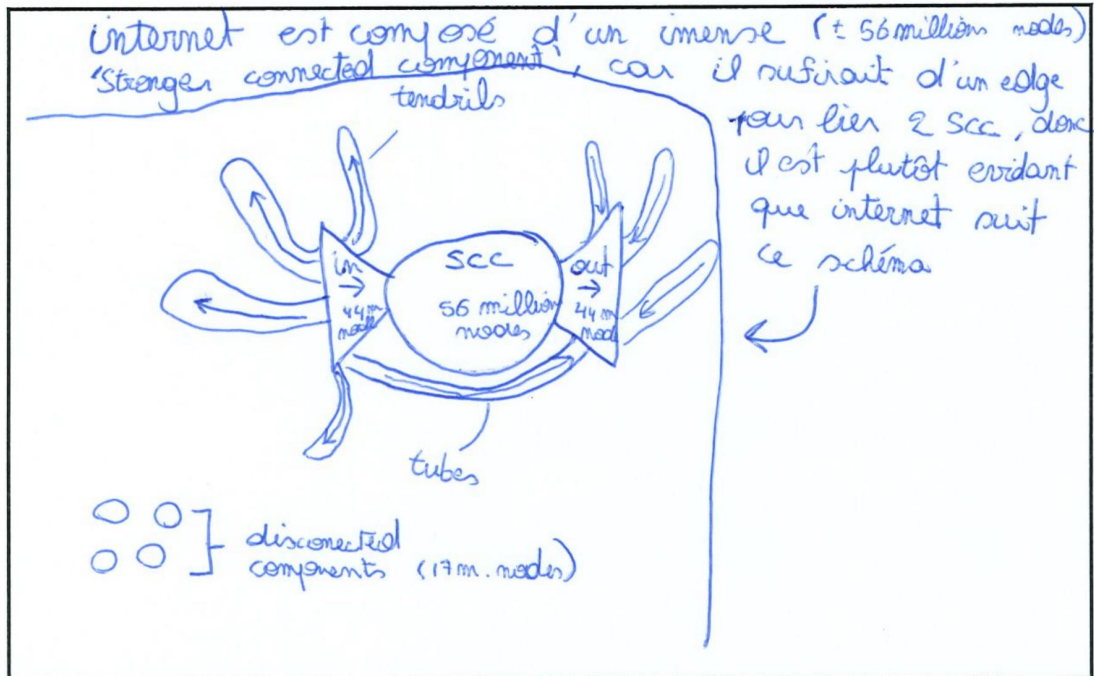
Define the concept of a balanced outcome of a network exchange. It is possible to prove that any network with a stable outcome also has a balanced outcome. Give an example of a balanced outcome in a four-node path. Show that it is balanced.



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Q4 Bow-tie structure of the Web [3pts]

Explain in detail the bow-tie structure of the Web, including the relative sizes of the different parts.



Q5 Bayes' Rule [4pts]

In a particular city, 90% of taxi cabs are black and 10% are yellow. A witness to an accident says the taxi was yellow. Assume witnesses sometimes misidentify colors of cabs: the color they say is only correct 75% of the time. Use Bayes' Rule to compute the probability that the cab is yellow, if the witness says it is yellow. Show the computation.

$$P(A|B) = \frac{P(A) \cdot P(B|A)}{P(B)} \quad \text{: Bayes' Rule}$$

$$P(Y) = 0.1, P(W_Y|Y) = 0.75, P(W_Y) = P(Y) \cdot P(W_Y|Y) + P(B) \cdot P(W_Y|B)$$

$$\Rightarrow P(Y|W_Y) = \frac{P(Y) \cdot P(W_Y|Y)}{P(W_Y)}$$

$$P(W_Y) = 0.1 \cdot 0.75 + 0.9 \cdot 0.25 = 0.3$$

donc

$$P(Y|W_Y) = \frac{0.1 \cdot 0.75}{0.3} = 0.25$$

si le témoin dit que le taxi est jaune, il y a 25% de chance qu'elle soit jaune.

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Q6 Direct-benefit effects [7pts]

Q6.1 Market without direct-benefit effects [3pts]

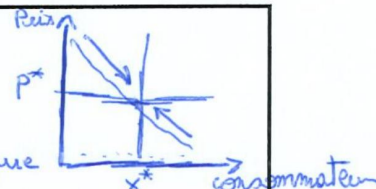
In the course we defined a simple continuous model of a market without direct-benefit effects. Consumers are given names x as real numbers from 0 to 1, uniformly distributed from 0 to 1. Each consumer has a reservation price $r(x)$, arranged in decreasing order. Using a graph, explain the relationship between the price of a product and the consumers who buy the product. Explain why the production cost p^* defines a *stable equilibrium* in this market with equilibrium price x^* . Why is it stable?

l'entreprise vend au moins à P^* sinon elle perd de l'argent.

c'est équilibré car pour tout x tel que $x < x^*$, $\pi(x) > \pi(x')$. Et inversement si $x > x^*$, $\pi(x) < \pi(x')$. ~~Les changements des prix~~

La pression sur les consommateurs fait de cet équilibre un équilibre stable.

C'est bien un équilibre stable car la zone sous la courbe est maximisée



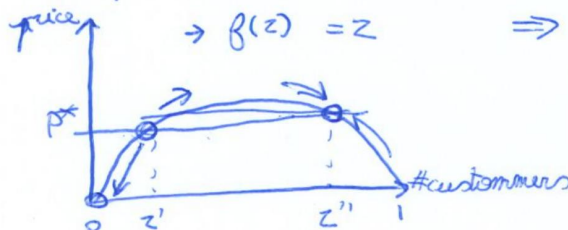
Q6.2 Market with direct-benefit effects [4pts]

Extend the model of the previous question to add direct-benefit effects. Add a second function $f(z)$ that quantifies the benefit to each consumer when a fraction z of all consumers has bought the product.

- What is the formula for the *new equilibrium price* in this extended market?
- Make simple assumptions about the form of the functions $r(x)$ and $f(z)$, as we did in the course. Given these assumptions, what is the *new graph* of consumers versus price?
- What are the *equilibria* in this new graph?
- Explain for each equilibrium whether it is *stable* or *unstable* and justify your explanations.

• Formule : $\pi(x) \cdot f(z)$.

• assumons que $\pi(x) = 1 - x \Rightarrow$ diminue entre 0 $\rightarrow$ 1.
 $\rightarrow f(z) = z \Rightarrow$ augmente entre 0 $\rightarrow$ 1.



ici il ya 3 équilibres $\{0, z', z''\}$, ~~est stable~~

- $z < z' \Rightarrow \pi(z) \cdot f(z) > \pi(z') \cdot f(z') \rightarrow$ ~~la~~ pression sur consommateurs $\rightarrow$ est stable. ~~faible~~

- $z' < z < z'' \Rightarrow \pi(z') \cdot f(z') > \pi(z) \cdot f(z) > \pi(z'') \cdot f(z'') \rightarrow$ ~~la~~ pression sur consommateurs $\rightarrow$ z' est instable

- $z'' < z \Rightarrow \pi(z'') \cdot f(z'') > \pi(z) \cdot f(z) \rightarrow$ ~~la~~ pression $\rightarrow$ z'' est stable